Problem Set 3

Applied Stats/Quant Methods 1

Due: November 12, 2026

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Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
model1 <- lm(voteshare ~ difflog, data = inc.sub)
```

Table 1: Regression of incumbent's vote share on difference in spending

	Dependent variable:
	Incumbent's vote share (voteshare)
Difference in spending (difflog)	0.042***
	(0.001)
Constant	0.579***
	(0.002)
Observations	3,193
R^2	0.367
Adjusted R^2	0.367
Residual Std. Error	0.079 (df = 3191)
F Statistic	1,852.791*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

The intercept (0.579) is the predicted vote share of the incumbent when `difflog` is 0, that is, when the incumbent and the challenger spend the same amount. In this case the incumbent is predicted to get about 57.9% of the vote. The slope (0.042) means that a one unit increase in `difflog` is associated with an increase of about 0.042 (4.2 percentage points) in the incumbent's vote share. The p-value of the slope is far below 0.05, so we reject the null hypothesis that $\beta_1 = 0$: the relationship is statistically significant. The R^2 shows that `difflog` explains about 36.7% of the variation in `voteshare`.

2. Make a scatterplot of the two variables and add the regression line.

```
plot(inc.sub$difflog, inc.sub$voteshare,
     xlab = "Difference in campaign spending (difflog)",
     ylab = "Incumbent's vote share",
     main = "Vote share and spending difference",
     pch = 16, cex = 0.5, col = "darkred")
abline(model1, col = "darkblue", lwd = 2)
```

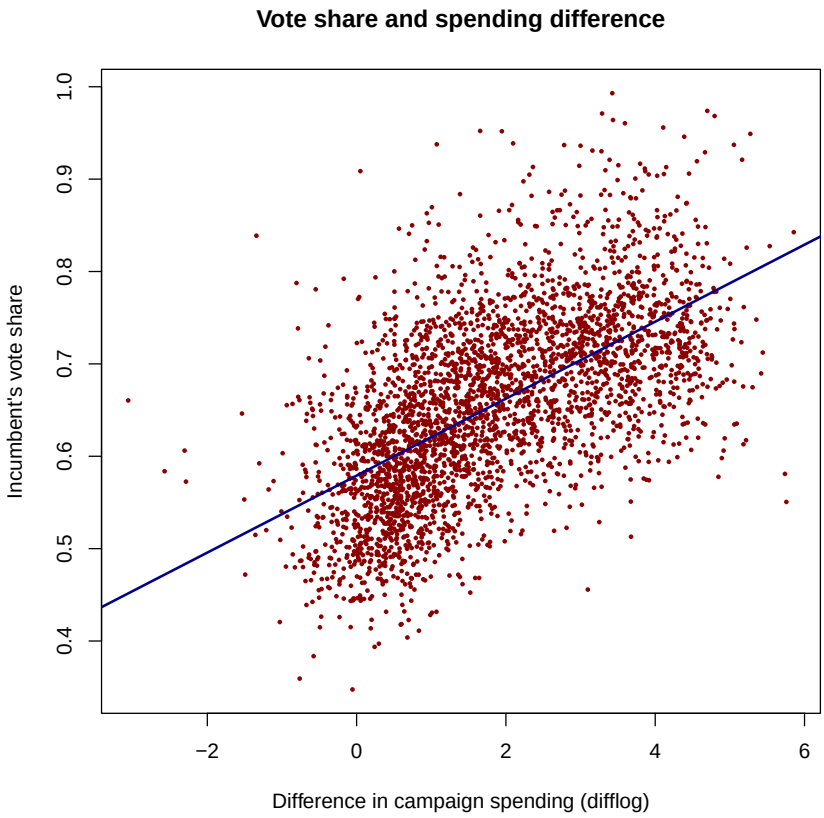


Figure 1: Scatterplot of `voteshare` and `difflog` with the regression line

The points and the line go upward: when the incumbent spends more than the challenger, the incumbent tends to get a higher vote share.

3. Save the residuals of the model in a separate object.

```
resid_1 <- residuals(model1)
```

Table 2: Summary of the residuals of Question 1 (resid_1)

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.2683	-0.0535	-0.0038	-0	0.0478	0.3275

Each residual is the difference between the observed vote share and the vote share predicted by the model, $\hat{\varepsilon}_i = y_i - \hat{y}_i$. They show the part of `voteshare` that is not explained by `difflog`. Their mean is 0, as always in OLS with an intercept.

4. Write the prediction equation.

$$\widehat{\texttt{voteshare}}_i = 0.579 + 0.042 \times \texttt{difflog}_i$$

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
model2 <- lm(presvote ~ difflog , data = inc.sub)
```

Table 3: Regression of presidential vote share on difference in spending

	Dependent variable:
	Vote share of the president's party (presvote)
Difference in spending (difflog)	0.024***
	(0.001)
Constant	0.508***
	(0.003)
Observations	3,193
R^2	0.088
Adjusted R^2	0.088
Residual Std. Error	0.110 (df = 3191)
F Statistic	307.715*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

When the two candidates spend the same amount (`difflog` = 0), the predicted vote share of the presidential candidate of the incumbent's party is about 50.8%. A one unit increase in `difflog` is associated with an increase of about 0.024 (2.4 percentage points) in `presvote`. The slope is statistically significant (p-value < 0.001), but the relationship is weaker than in Question 1: `difflog` explains only about 8.8% of the variation in `presvote`.

2. Make a scatterplot of the two variables and add the regression line.

```r
plot(inc.sub$difflog, inc.sub$presvote,
     xlab = "Difference in campaign spending (difflog)",
     ylab = "Vote share of the president's party (presvote)",
     main = "Presidential vote share and spending difference",
     pch = 16, cex = 0.5, col = "darkred")
abline(model2, col = "darkblue", lwd = 2)
```

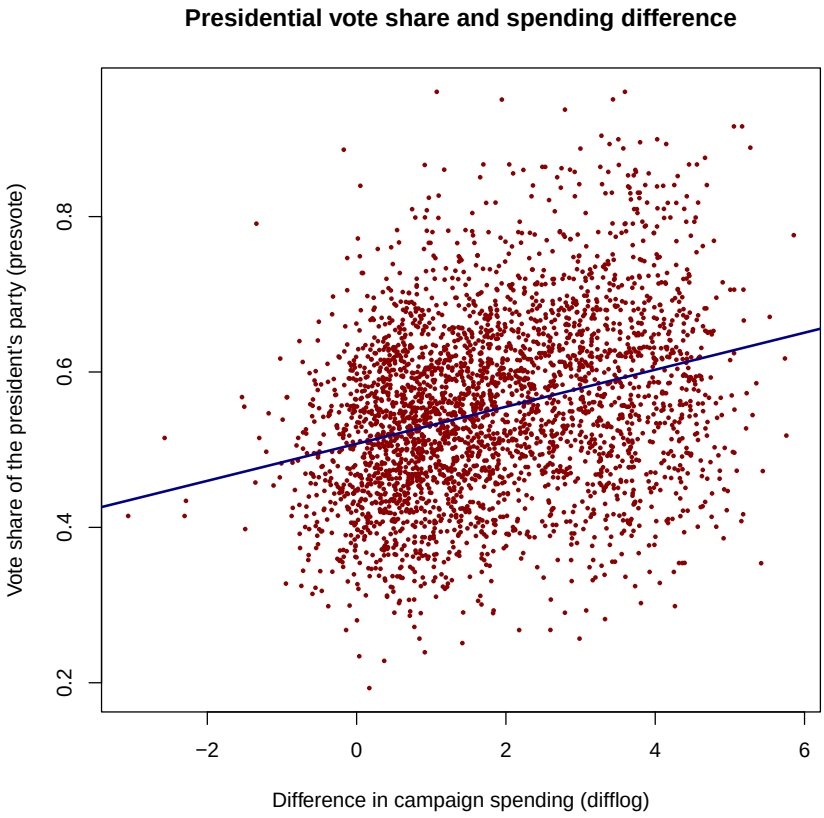


Figure 2: Scatterplot of `presvote` and `difflog` with the regression line

The line goes slightly upward, but the points are much more spread around it than in Question 1, which matches the low R^2.

3. Save the residuals of the model in a separate object.

```r
resid_2 <- residuals(model2)
```

Table 4: Summary of the residuals of Question 2 (resid_2)

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.3220	-0.0741	-0.0010	0	0.0715	0.4274

These residuals show the part of `presvote` that is not explained by `difflog`.

4. Write the prediction equation.

$$\widehat{\texttt{presvote}}_i = 0.508 + 0.024 \times \texttt{difflog}_i$$

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
model3 <- lm(voteshare ~ presvote, data = inc.sub)
```

Table 5: Regression of incumbent's vote share on presidential vote share

	Dependent variable:
	Incumbent's vote share (voteshare)
Vote share of the president's party (presvote)	0.388***
	(0.013)
Constant	0.441***
	(0.008)
Observations	3,193
R^2	0.206
Adjusted R^2	0.206
Residual Std. Error	0.088 (df = 3191)
F Statistic	826.950*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

The slope (0.388) means that when the vote share of the presidential candidate of the incumbent's party increases by 1 percentage point, the incumbent's vote share

is predicted to increase by about 0.39 percentage points. The slope is statistically significant (p-value < 0.001), and `presvote` explains about 20.6% of the variation in `voteshare`. The intercept (0.441) is the predicted vote share when `presvote` is 0. This value is outside the range of the data, so it has no useful meaning on its own.

2. Make a scatterplot of the two variables and add the regression line.

```
plot(inc.sub$presvote, inc.sub$voteshare,
     xlab = "Vote share of the president's party (presvote)",
     ylab = "Incumbent's vote share (voteshare)",
     main = "Incumbent vote share and presidential vote share",
     pch = 16, cex = 0.5, col = "darkred")
abline(model3, col = "darkblue", lwd = 2)
```

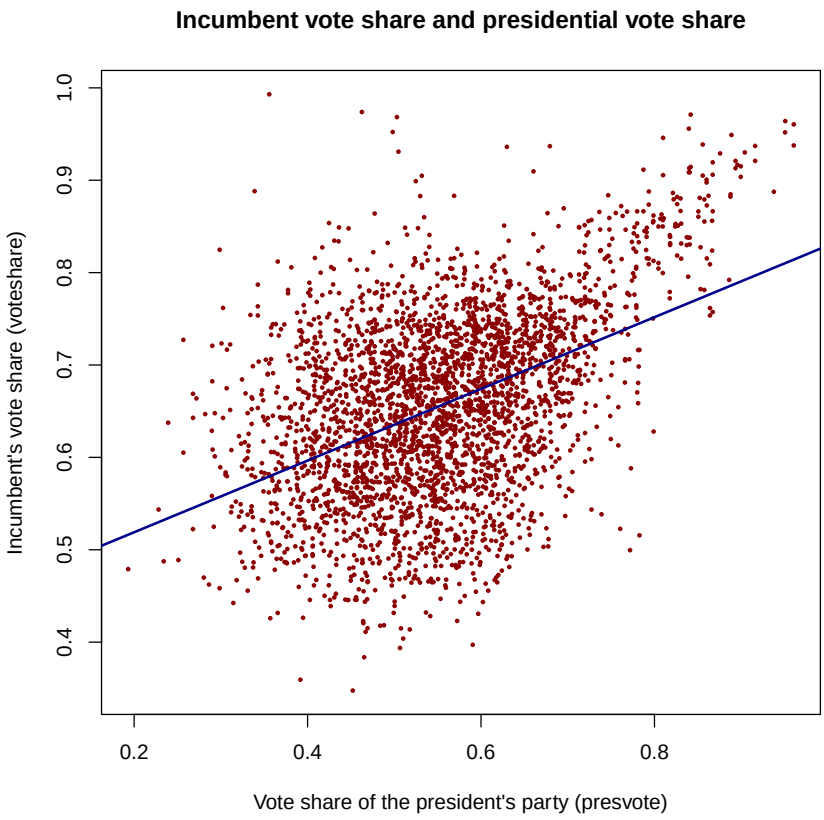


Figure 3: Scatterplot of `voteshare` and `presvote` with the regression line

The plot shows a positive relationship: incumbents tend to do better in districts where the presidential candidate of their party also does better.

3. Write the prediction equation.

$$\widehat{\texttt{voteshare}}_i = 0.441 + 0.388 \times \texttt{presvote}_i$$

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
model4 <- lm(resid_1 ~ resid_2)
```

Table 6: Regression of the residuals of Question 1 on the residuals of Question 2

	Dependent variable:
	Residuals of Question 1 (resid_1)
Residuals of Question 2 (resid_2)	0.257***
	(0.012)
Constant	−0.000
	(0.001)
Observations	3,193
R^2	0.130
Adjusted R^2	0.130
Residual Std. Error	0.073 (df = 3191)
F Statistic	476.975*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

The intercept is 0 because both sets of residuals have a mean of 0. The slope (0.257) shows the relationship between `presvote` and `voteshare` after the effect of `difflog` has been removed from both variables. It is smaller than the slope in Question 3 (0.388), because part of the relationship between `presvote` and `voteshare` was due to their common relationship with `difflog`. The slope is still statistically significant (p-value < 0.001).

2. Make a scatterplot of the two residuals and add the regression line.

```r
plot(resid_2, resid_1,
     xlab = "presvote not explained by difflog",
     ylab = "voteshare not explained by difflog",
     main = "Residuals of Problem 1 and Problem 2",
     pch = 16, cex = 0.5, col = "darkred")
abline(model4, col = "darkblue", lwd = 2)
```

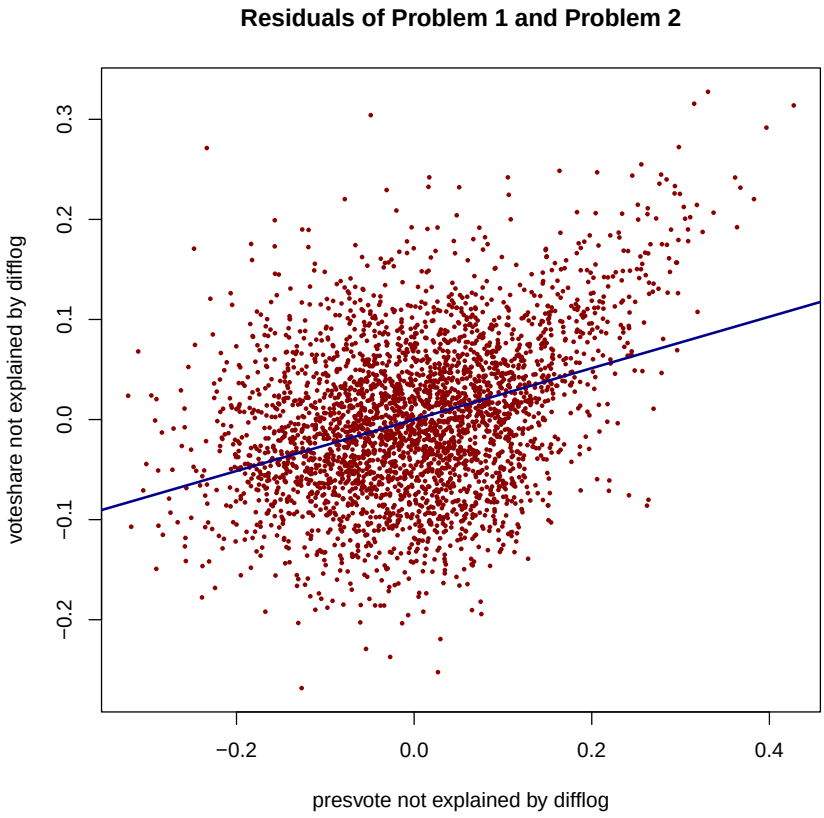


Figure 4: Added variable plot: residuals of Question 1 against residuals of Question 2

This is the added variable plot. The points are centred around $(0, 0)$ and the line goes upward, showing that `presvote` is still positively related to `voteshare` once `difflog` is taken into account.

3. Write the prediction equation.

$$\widehat{\texttt{resid_1}}_i = 0 + 0.257 \times \texttt{resid_2}_i$$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
model5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
```

Table 7: Regression of incumbent's vote share on difference in spending and presidential vote share

	Dependent variable:
	Incumbent's vote share (voteshare)
Difference in spending (difflog)	0.036***
	(0.001)
Vote share of the president's party (presvote)	0.257***
	(0.012)
Constant	0.449***
	(0.006)
Observations	3,193
R^2	0.450
Adjusted R^2	0.449
Residual Std. Error	0.073 (df = 3190)
F Statistic	1,302.947*** (df = 2; 3190)
Note:	*p<0.1; **p<0.05; ***p<0.01

Holding `presvote` constant, a one unit increase in `difflog` is associated with an increase of about 0.036 in the incumbent's vote share. Holding `difflog` constant, a 1 percentage point increase in `presvote` is associated with an increase of about 0.26 percentage points in the incumbent's vote share. Both coefficients are statistically significant (p-value < 0.001), and together the two variables explain about 45% of the variation in `voteshare`.

2. Write the prediction equation.

$$\widehat{\texttt{voteshare}}_i = 0.449 + 0.036 \times \texttt{difflog}_i + 0.257 \times \texttt{presvote}_i$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

   The coefficient of `presvote` in Question 5 is identical to the slope in Question 4: both are 0.2569, with the same standard error (0.0118) and the same t value (21.84).

   This happens because of how multiple regression works. The residuals from Question 1 are the part of `voteshare` that is not explained by `difflog`, and the residuals from Question 2 are the part of `presvote` that is not explained by `difflog`. When we regress one set of residuals on the other, we measure the relationship between `presvote` and `voteshare` after removing the effect of `difflog` from both. This is exactly what the coefficient of `presvote` measures in the multiple regression: the effect of `presvote` holding `difflog` constant. This result is known as the Frisch-Waugh-Lovell theorem, and the plot in Question 4 is the added variable plot for `presvote`. The residual standard errors are only very slightly different (0.07338 and 0.07339) because Question 5 estimates one more coefficient, so it has one less degree of freedom (3190 instead of 3191).